Results in Control and Optimization 4 (2021) 100036

Contents lists available at ScienceDirect

Results in Control and Optimization

journal homepage: www.elsevier.com/locate/rico

Some new rearrangements in sensitivity integrals and concerning inequalities with their application in control

Shaival H. Nagarsheth *, Shambhu N. Sharma

Electrical Engineering Department, Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat, Gujarat 395007, India

ARTICLE INFO

Keywords:
Sensitivity functions
Logarithmic sensitivity integral
Logarithmic sensitivity inequality
Loop robustness

ABSTRACT

In this paper, we extend the sensitivity integrals for linear feedback systems via a non-analytic sensitivity function in Poisson–Jensen's terminology. First, we sketch a generalized sensitivity integral accounting for the NMP zeros and the RHP poles of the sensitivity function. Interestingly, the sensitivity integral accounting for the NMP zeros becomes a special case of the generalized integral. Then the concerning sensitivity inequality and the associated lower bound are achieved. We also sketch a sensitivity integral that exploits a modified sensitivity function-based approach and stabilizes the linear feedback system. Their concerning sensitivity inequality and the associated lower bound are achieved as well. An illustrative case study example is employed to demonstrate the idea of the 'sensitivity lower bound' to choose the ability of controllers for the given linear feedback systems. The rearranged sensitivity lower bound offers simplified analysis involving fewer computation efforts.

1. Introduction

The role of sensitivity function is attributed to the ubiquity of control systems in diverse field. The sensitivity function has found applications in the study of qualitative characteristics of linear feedback systems subject to variation in parameters, stability margin as well as loop robustness [1]. The goal in the design of the controller is not only to achieve the desired output but also loop robustness. Zames and Francis [2] have explained the usefulness of the sensitivity function. The importance of sensitivity function is directly linked to disturbance rejection, small parameter variations, and tracking errors [3]. Rolf Nevanlinna introduced the Poisson–Jensen formula, which is a direct consequence of Jensen's formula [4], by applying the Möbius transformation in the discrete domain. Subsequently, Levinson and Redheffer [5] proved the same result in continuous settings as a particular case [6]. Freudenberg and Looze [7] represented the Poisson–Jensen integral as a constraint on the feedback design of control systems. H. W. Bode [8] studied the constraints on the linear time-invariant feedback system with single input single output settings. The design trade-offs and limitations of closed-loop feedback systems can be expressed in terms of the sensitivity function. Bode's Theorem states that for the stable open-loop transfer function, the log magnitude of the sensitivity function must equal zero. In the presence of NMP zeros and RHP poles, the result of Bode's Theorem changes and the integral constraints vary [9]. A similar concept has been extended to linear multivariable feedback systems [10–12]. The Bode integral was extended to linear time-varying systems [13] as well as linear time-periodic [14]. This integral relation states that if the disturbance attenuation is improved in the sense of the sensitivity reduction in a frequency range, it worsens in another frequency range. This property of the constraint is described as a waterbed effect [1]. Recently, an advancement in the solution of the 30 years old sensitivity conjecture has been provided [15]. Saravanakumar et al. [16] developed a dissipative optimal control scheme utilizing Jensen's integral inequality to deal with input nonlinearities and randomly occurring perturbations; see [17] as well for an optimal control scheme to deal with uncertain parameters for the switched neutral

* Corresponding author.
E-mail addresses: shn411@gmail.com (S.H. Nagarsheth), snsvolterra@gmail.com (S.N. Sharma).

https://doi.org/10.1016/j.rico.2021.100036
Received 1 February 2021; Received in revised form 14 May 2021; Accepted 27 June 2021
Available online 7 July 2021
2666-7207/© 2021 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

Notations

$G(s)$	Forward-path transfer function
$\widetilde{G}(s)$	Modified forward-path transfer function
$G_p(s)$	Plant transfer function
$G_c(s)$	Controller transfer function
$g(s)$	Sensitivity function
$\widetilde{g}(s)$	Modified sensitivity function
α_i	ith NMP zero associated with sensitivity function
β_i	ith RHP pole associated with sensitivity function
$\sigma_k + j\eta_k$	Singular point in the RH
ς_k	kth NMP zero associated with sensitivity function
s_{lb}	Sensitivity lower bound
$s_{\max}$	Absolute maximum sensitivity
ρ	Absolute minimum of the sensitivity function
ω_c	Crossing frequency
k_p	System gain
k_c	Controller gain
t_d	Dead-time
τ_1, τ_2	Time constant
τ_I	Integral time constant
τ_d	Derivative time constant
a	Variable gain
NMP	Non-Minimum Phase
RHP	Right Half-Plane

systems. In [18] non-fragile optimal controllers are proposed for Markovian jump systems under randomly occurring perturbations and time-varying delay. A good source of stochastic fault-tolerant control for random uncertainties, stochastic failures and external disturbances can be found in [19].

Satisfactory results are available based on sensitivity function for linear processes [20] as well as nonlinear processes [21]. Ringwood et al. [22] can be referred for sensitivity analysis of model-based wave energy converter control schemes. In recent years, several PID tunings rules [23,24], fractional-order controllers [25,26], Model Predictive Control (MPC) [27,28], model-driven compensation techniques [29], linear-filter design [30] are presented based on the sensitivity approach. However, few researchers have attempted to apply sensitivity integrals to practical problems [31,32]. Stein [8] unfolded sensitivity integrals for aerodynamic problems. The sensitivity aspect of addressing the process uncertainty in the ship steering system is available in Roberts & Towill [33]. Controller tuning achieving robustness to address the uncertainty makes a control problem hard [34]. Due to the inadequacy of the controller (lack of loop robustness) to cope up with the input disturbance and process parameter variations, the closed-loop linear feedback systems are pulled into instability. It is imperative to resolve the problem of stabilizing linear feedback systems that are pulled into instability attributed to uncertainty and input disturbances. The analysis from the sensitivity integral perspective to examine the controllers' ability is achieved chiefly in this paper.

This paper revisits the sensitivity integral to address the problem of examining the ability of linear feedback systems to handle the effect of input disturbances and uncertainties. We utilize the non-analytic sensitivity function theory combined with the Poisson–Jensen type sensitivity integrals and Cauchy residue integration theory to arrive at the results of this paper. Theorem 1 extends the conventional sensitivity integral, and its related inequality is derived in Theorem 2. The inequality is rearranged, resulting in fewer computational efforts. The problem of augmenting the stabilization boils down to the problem of 'reshaping the forward path' transfer function and hence 'reshaping the sensitivity inequality'. Theorems 3 and 4 are entrusted the task of unifying the idea of the Poisson–Jensen-type complex function integral with the modified sensitivity function to address the problem of augmenting the stabilization. The usefulness of the proposed sensitivity lower bound is demonstrated by employing a case study example. The usefulness of the proposed procedure is that the robustness of the controller can be adjudged using the sensitivity lower bound without dwelling into the computational subtitles associated with the calculation of absolute maximum sensitivity.

This paper is organized as follows: Section 2 consists of the main theoretical results with four theorems and their systematic proofs related to the logarithmic sensitivity integrals and their inequalities. Section 3 presents a case study example, which demonstrates the usefulness of the main results of this paper. The conclusion is given in Section 4.

2. Main results

To achieve the objective of this paper, first, the non-analytic function theory is used to examine the ability of linear feedback systems to handle the effect of input disturbances and process parameter variations. The related logarithmic integrals and inequalities

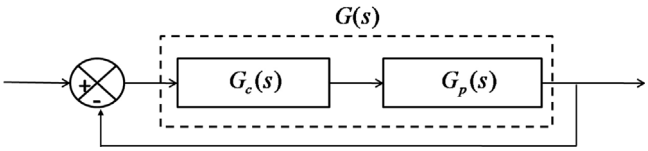

Fig. 1. Linear feedback system.

are formalized in Theorems 1 and 2. Then the analytic function theory is used to augment the ability to handle the effect of input disturbances and process parameter variations, which can be found in Theorems 3 and 4 of this paper.

2.1. Sensitivity integrals and related inequalities

Fig. 1 shows the block diagram of a linear feedback system taken into consideration. Here $G_p(s)$ is a process transfer function, $G_c(s)$ is the controller transfer function and $G(s)$ is the forward path transfer function of the feedback loop.

The sensitivity function $g(s)$ is defined as

$$g(s) = \frac{1}{1+G(s)}, \tag{1}$$

where $G(s) = G_c(s)G_p(s)$. In the structure of sensitivity function $g(s)$ the NMP zeros of $g(s)$ can be regarded as unstable open-loop poles. The poles of the function $g(s)$ become the closed-loop poles of the linear feedback system. To cover a broader class of cases, we consider the non-analytic sensitivity function consisting of non-minimum phase zeros and right half plane poles. The sensitivity integral of the sensitivity function gives a trajectory of the absolute sensitivity over a range of frequencies for the linear feedback system. That reveals the attenuation as well as the amplification area of the disturbances and uncertainties.

Theorem 1 invokes the idea of weaving the Poisson–Jensen type sensitivity integral to formalize the instability arising in the linear feedback system. After combining the idea from the non-analytic function and the Cauchy residue integration theory, we rearrange the associated sensitivity function and the concerning sensitivity lower bound in Theorems 1 and 2, respectively. That can be achieved by first designing a compensator resulting from the Blaschke product and the sensitivity function. Then, thanks to the Cauchy Residue complex integration Theorem, we construct the Cauchy integral for logarithmic modified sensitivity function.

Theorem 1. *Consider a feedback control loop with the sensitivity function $g(s)$, which has the following properties:*

(i) the function $g(s)$ is non-analytic
(ii) the sensitivity function $g(s)$ has non-minimum phase zeros $\alpha_1, \alpha_2, \dots, \alpha_M$ in the open Right Half Plane (RHP)
(iii) the sensitivity function $g(s)$ has $\beta_1, \beta_2, \dots, \beta_N$ RHP poles.

Then, the Poisson–Jensen formula becomes

$$\int_{-\infty}^{\infty} \log|g(j\omega)| \frac{\sigma_k}{\sigma_k^2+(\omega-\eta_k)^2} d\omega = -\pi(-\log|g(\sigma_k+j\eta_k)| + \log \sum_{1\le i\le M} \log\left|\frac{\sigma_k+j\eta_k-\alpha_i}{\sigma_k+j\eta_k+\overline{\alpha}_i}\right| + \sum_{1\le i\le N} \log\left|\frac{\sigma_k+j\eta_k+\overline{\beta}_i}{\sigma_k+j\eta_k-\beta_i}\right|),$$

where $\sigma_k + j\eta_k$ is an arbitrary singular point in the RHP.

Suppose the open-loop transfer function $G(s)$ associated with the sensitivity function $g(s)$ has the NMP zeros. Consider the singular point $\sigma_k + j\eta_k = \varsigma_k$, where ς_k is the kth NMP open-loop zero associated with the sensitivity function $g(s)$. Then, the Poisson–Jensen formula becomes

$$\int_{-\infty}^{\infty} \log|g(j\omega)| \frac{\sigma_k}{\sigma_k^2+(\omega-\eta_k)^2} d\omega = -\pi(\sum_{1\le i\le M} \log\left|\frac{\sigma_k+j\eta_k-\alpha_i}{\sigma_k+j\eta_k+\overline{\alpha}_i}\right| + \sum_{1\le i\le N} \log\left|\frac{\sigma_k+j\eta_k+\overline{\beta}_i}{\sigma_k+j\eta_k-\beta_i}\right|),$$

where $1 \le k \le L$ and L is restricted to the number of NMP open-loop zeros.

Proof. We associate a singular point s_0 with the sensitivity function $g(s)$. The singular point s_0 is arbitrary or coincides with one of the NMP zeros of the open-loop transfers function $G(s)$. After associating the singular point, i.e., lying in the RHP, with the logarithmic function $\log g(s)$, the non-analyticity arises. The NMP zeros of the sensitivity function contained in the open RHP does not introduce non-analyticity, but the logarithmic sensitivity function has non-analyticity. We construct a new function $\widetilde{g}(s)$ with the analyticity, i.e.

$$\widetilde{g}(s) = g(s) \prod_{1\le i\le M} \left(\frac{s+\overline{\alpha}_i}{s-\alpha_i}\right) \prod_{1\le i\le N} \left(\frac{s-\beta_i}{s+\overline{\beta}_i}\right), \quad \frac{\widetilde{g}(s)}{s-s_0} = g(s)\frac{\prod_{1\le i\le M}\left(\frac{s+\overline{\alpha}_i}{s-\alpha_i}\right)\prod_{1\le i\le N}\left(\frac{s-\beta_i}{s+\overline{\beta}_i}\right)}{s-s_0}, \tag{2a}$$

has the analyticity and non-analyticity, respectively. From the Cauchy integration Theorem [5], we have

$$\log \widetilde{g}(s_0) = -\frac{1}{2\pi j}\oint \frac{\log \widetilde{g}(s)}{s-s_0} ds + \frac{1}{2\pi j}\oint \frac{\log \widetilde{g}(s)}{s-s_1}. \tag{2b}$$

Note that the point s_1, contained in the left half-plane is an image of the singular point s_0. The function $\frac{\log \widetilde{g}(s)}{s-s_1}$ has the analyticity, and the function $\frac{\log \widetilde{g}(s)}{s-s_0}$ has non-analyticity in the right half-plane [10]. Furthermore,

$$\log \widetilde{g}(s_0) = \log g(s_0) + \sum_{1\le i\le M} \log \frac{s_0+\overline{\alpha}_i}{s_0-\alpha_i} + \sum_{1\le i\le N} \log \frac{s_0-\beta_i}{s_0+\overline{\beta}_i} = -\frac{1}{2\pi j}\oint \frac{(s_0-s_1)\log \widetilde{g}(s)}{(s-s_0)(s-s_1)}ds. \tag{3}$$

Eq. (3) is a consequence of (1) and (2) that can be further recast as

$$\log g(s_0) + \sum_{1\le i\le M} \log \frac{s_0+\overline{\alpha}_i}{s_0-\alpha_i} + \sum_{1\le i\le N} \log \frac{s_0-\beta_i}{s_0+\overline{\beta}_i} = -\frac{1}{2\pi j}\oint \frac{(s_0-s_1)}{(s-s_0)(s-s_1)}(\log g(s) + \sum_{1\le i\le M} \log \frac{s+\overline{\alpha}_i}{s-\alpha_i} + \sum_{1\le i\le N} \log \frac{s-\beta_i}{s+\overline{\beta}_i}).$$

Thus,

$$\begin{aligned}\log g(s_0) + \sum_{1\le i\le M} \log \frac{s_0+\overline{\alpha}_i}{s_0-\alpha_i} + \sum_{1\le i\le N} \log \frac{s_0-\beta_i}{s_0+\overline{\beta}_i} &= -\frac{1}{2\pi j}\oint \frac{(s_0-s_1)\log g(s)}{(s-s_0)(s-s_1)}ds \\ &- \frac{1}{2\pi j}\sum_{1\le i\le M}\oint \frac{s_0-s_1}{(s-s_0)(s-s_1)}\log\frac{s+\overline{\alpha}_i}{s-\alpha_i}ds \\ &- \frac{1}{2\pi j}\sum_{1\le i\le N}\oint \frac{s_0-s_1}{(s-s_0)(s-s_1)}\log\frac{s-\beta_i}{s+\overline{\beta}_i}ds.\end{aligned} \tag{4}$$

After decomposing integrals of (4) into parts, we arrive at

$$\begin{aligned}\log g(s_0) + \sum_{1\le i\le M} \log \frac{s_0+\overline{\alpha}_i}{s_0-\alpha_i} + \sum_{1\le i\le N} \log \frac{s_0-\beta_i}{s_0+\overline{\beta}_i} &= -\frac{1}{2\pi j}\oint \frac{(s_0-s_1)\log g(s)}{(s-s_0)(s-s_1)}ds \\ &- \frac{1}{2\pi j}\sum_{1\le i\le M}\oint \frac{s_0-s_1}{(s-s_0)(s-s_1)}\log\frac{s+\overline{\alpha}_i}{s-\alpha_i}ds \\ &- \frac{1}{2\pi j}\sum_{1\le i\le N}\oint \frac{s_0-s_1}{(s-s_0)(s-s_1)}\log\frac{s-\beta_i}{s+\overline{\beta}_i}ds.\end{aligned}$$

Furthermore,

$$\begin{aligned}\log g(s_0) + \sum_{1\le i\le M} \log \frac{s_0+\overline{\alpha}_i}{s_0-\alpha_i} + \sum_{1\le i\le N} \log \frac{s_0-\beta_i}{s_0+\overline{\beta}_i} &= -\frac{1}{2\pi j}\int_{-\infty}^{\infty} \frac{j(s_0-s_1)\log g(j\omega)}{(j\omega-s_0)(j\omega-s_1)}d\omega \\ &- \frac{1}{2\pi j}\underset{R\to\infty}{Lt}\int_{\frac{\pi}{2}}^{-\frac{\pi}{2}} \frac{(s_0-s_1)\log g(\mathrm{Re}^{j\gamma})}{(\mathrm{Re}^{j\gamma}-s_0)(\mathrm{Re}^{j\lambda}-s_1)}\mathrm{Re}^{j\gamma}jd\gamma \\ &- \frac{1}{2\pi j}\sum_{1\le i\le M}\oint \frac{s_0-s_1}{(s-s_0)(s-s_1)}\log\frac{s+\overline{\alpha}_i}{s-\alpha_i}ds \\ &- \frac{1}{2\pi j}\sum_{1\le i\le N}\oint \frac{s_0-s_1}{(s-s_0)(s-s_1)}\log\frac{s-\beta_i}{s+\overline{\beta}_i}ds.\end{aligned}$$

Note that an open RHP can be regarded as a semi-circle with the radius $R\to\infty$ and

$$\underset{R\to\infty}{Lt}\frac{\log g(\mathrm{Re}^{j\gamma})}{(\mathrm{Re}^{j\gamma}-s_0)(\mathrm{Re}^{j\gamma}-s_1)}\mathrm{Re}^{j\gamma} = \underset{R\to\infty}{Lt}\frac{-\log(1+G(\mathrm{Re}^{j\gamma}))}{(\mathrm{Re}^{j\gamma}-s_0)(\mathrm{Re}^{j\gamma}-s_1)}\mathrm{Re}^{j\gamma} = 0.$$

for every relative degree of the open-loop transfer function $G(s)$.

Thus,

$$\begin{aligned}\log g(s_0) + \sum_{1\le i\le M} \log \frac{s_0+\overline{\alpha}_i}{s_0-\alpha_i} + \sum_{1\le i\le N} \log \frac{s_0-\beta_i}{s_0+\overline{\beta}_i} &= -\frac{1}{2\pi j}\int_{-\infty}^{\infty} \frac{j(s_0-s_1)\log g(j\omega)}{(j\omega-s_0)(j\omega-s_1)}d\omega \\ &- \frac{1}{2\pi j}\sum_{1\le i\le M}\oint \frac{s_0-s_1}{(s-s_0)(s-s_1)}\log\frac{s+\overline{\alpha}_i}{s-\alpha_i}ds \\ &- \frac{1}{2\pi j}\sum_{1\le i\le N}\oint \frac{s_0-s_1}{(s-s_0)(s-s_1)}\log\frac{s-\beta_i}{s+\overline{\beta}_i}ds.\end{aligned} \tag{5}$$

Furthermore,

$$\begin{aligned}-\frac{1}{2\pi j}\sum_{1\le i\le M}\oint \frac{s_0-s_1}{(s-s_0)(s-s_1)}\log\frac{s+\overline{\alpha}_i}{s-\alpha_i}ds &= -\frac{1}{2\pi j}\sum_{1\le i\le M}\int_{-\infty}^{\infty} \frac{j(s_0-s_1)}{(j\omega-s_0)(j\omega-s_1)}\log\frac{j\omega+\overline{\alpha}_i}{j\omega-\alpha_i}d\omega \\ &- \frac{1}{2\pi j}\sum_{1\le i\le M}\underset{R\to\infty}{Lt}\int_{\frac{\pi}{2}}^{-\frac{\pi}{2}} \frac{(s_0-s_1)}{(\mathrm{Re}^{j\gamma}-s_0)(\mathrm{Re}^{j\gamma}-s_1)}\log(\frac{\mathrm{Re}^{j\gamma}+\overline{\alpha}_i}{\mathrm{Re}^{j\gamma}-\alpha_i})\mathrm{Re}^{j\gamma}jd\gamma \\ &= -\frac{1}{2\pi j}\sum_{1\le i\le M}\int_{-\infty}^{\infty} \frac{j(s_0-s_1)}{(j\omega-s_0)(j\omega-s_1)}\log\frac{j\omega+\overline{\alpha}_i}{j\omega-\alpha_i}d\omega\end{aligned}$$

Under the limiting case, the second term of the right-hand side of the above vanishes. Thus,

$$-\frac{1}{2\pi j}\sum_{1\le i\le M}\oint \frac{s_0-s_1}{(s-s_0)(s-s_1)}\log\frac{s+\overline{\alpha}_i}{s-\alpha_i}ds=-\frac{1}{2\pi j}\sum_{1\le i\le M}\int_{-\infty}^{\infty}\frac{j(s_0-s_1)}{(j\omega-s_0)(j\omega-s_1)}\log\frac{j\omega+\overline{\alpha}_i}{j\omega-\alpha_i}d\omega. \tag{6a}$$

The term

$$-\frac{1}{2\pi j}\sum_{1\le i\le N}\oint \frac{s_0-s_1}{(s-s_0)(s-s_1)}\log\frac{s-\beta_i}{s+\overline{\beta}_i}ds=-\frac{1}{2\pi j}\sum_{1\le i\le N}\int_{-\infty}^{\infty}\frac{j(s_0-s_1)}{(j\omega-s_0)(j\omega-s_1)}\log\frac{j\omega-\beta_i}{j\omega+\overline{\beta}_i}d\omega$$
$$-\frac{1}{2\pi j}\sum_{1\le i\le N}\underset{R\to\infty}{Lt}\int_{\frac{\pi}{2}}^{-\frac{\pi}{2}}\frac{(s_0-s_1)}{(\mathrm{Re}^{j\gamma}-s_0)(\mathrm{Re}^{j\gamma}-s_1)}\log(\frac{\mathrm{Re}^{j\gamma}-\beta_i}{\mathrm{Re}^{j\gamma}+\overline{\beta}_i})\mathrm{Re}^{j\gamma}jd\gamma.$$

Under the limiting case, the second term of the right-hand side of the above vanishes as well. Thus

$$-\frac{1}{2\pi j}\sum_{1\le i\le M}\oint \frac{s_0-s_1}{(s-s_0)(s-s_1)}\log\frac{s+\overline{\alpha}_i}{s-\alpha_i}ds=-\frac{1}{2\pi j}\sum_{1\le i\le N}\int_{-\infty}^{\infty}\frac{j(s_0-s_1)}{(j\omega-s_0)(j\omega-s_1)}\log\frac{j\omega-\beta_i}{j\omega+\overline{\beta}_i}d\omega. \tag{6b}$$

After combining (5)–(6b), we get

$$\log g(s_0)+\sum_{1\le i\le M}\log\frac{s_0+\overline{\alpha}_i}{s_0-\alpha_i}+\sum_{1\le i\le N}\log\frac{s_0-\beta_i}{s_0+\overline{\beta}_i}=\frac{1}{2\pi j}\int_{-\infty}^{\infty}\frac{j(s_0-s_1)\log g(j\omega)}{(j\omega-s_0)(j\omega-s_1)}d\omega$$
$$-\frac{1}{2\pi j}\sum_{1\le i\le M}\int_{-\infty}^{\infty}\frac{j(s_0-s_1)}{(j\omega-s_0)(j\omega-s_1)}\log\frac{j\omega+\overline{\alpha}_i}{j\omega-\alpha_i}d\omega$$
$$-\frac{1}{2\pi j}\sum_{1\le i\le N}\int_{-\infty}^{\infty}\frac{j(s_0-s_1)}{(j\omega-s_0)(j\omega-s_1)}\log\frac{j\omega-\beta_i}{j\omega+\overline{\beta}_i}d\omega. \tag{7}$$

The idea is to simplify (7) without approximating the expression. Here, we consider $s_0=\sigma_k+j\eta_k, s_1=-\sigma_k+j\eta_k$ as well as account for the real parts of both sides. As a result of this, we are led to a more straightforward expression, i.e.

$$\mathrm{Re}(\log g(\sigma_k+j\eta_k)+\sum_{1\le i\le M}\log\frac{\sigma_k+j\eta_k+\overline{\alpha}_i}{\sigma_k+j\eta_k-\alpha_i}+\sum_{1\le i\le N}\log\frac{\sigma_k+j\eta_k-\beta_i}{\sigma_k+j\eta_k+\overline{\beta}_i})=\mathrm{Re}(\frac{1}{\pi}\int_{-\infty}^{\infty}\frac{\sigma_k\log g(j\omega)}{\sigma_k^2+(\omega-\eta_k)^2}d\omega$$
$$+\frac{1}{\pi}\sum_{1\le i\le M}\int_{-\infty}^{\infty}\frac{\sigma_k}{\sigma_k^2+(\omega-\eta_k)^2}\log\frac{j\omega+\overline{\alpha}_i}{j\omega-\alpha_i}d\omega+\frac{1}{\pi}\sum_{1\le i\le N}\int_{-\infty}^{\infty}\frac{\sigma_k}{\sigma_k^2+(\omega-\eta_k)^2}\log\frac{j\omega-\beta_i}{j\omega+\overline{\beta}_i}d\omega). \tag{8a}$$

Thus,

$$\log\left|g(\sigma_k+j\eta_k)\right|+\sum_{1\le i\le M}\log\left|\frac{\sigma_k+j\eta_k+\overline{\alpha}_i}{\sigma_k+j\eta_k-\alpha_i}\right|+\sum_{1\le i\le N}\log\left|\frac{\sigma_k+j\eta_k-\beta_i}{\sigma_k+j\eta_k+\overline{\beta}_i}\right|=\frac{1}{\pi}\int_{-\infty}^{\infty}\frac{\sigma_k\log|g(j\omega)|}{\sigma_k^2+(\omega-\eta_k)^2}d\omega$$
$$+\frac{1}{\pi}\sum_{1\le i\le N}\int_{-\infty}^{\infty}\frac{\sigma_k}{\sigma_k^2+(\omega-\eta_k)^2}\log\left|\frac{j\omega+\overline{\alpha}_i}{j\omega-\alpha_i}\right|d\omega$$
$$+\frac{1}{\pi}\sum_{1\le i\le M}\int_{-\infty}^{\infty}\frac{\sigma_k}{\sigma_k^2+(\omega-\eta_k)^2}\log\left|\frac{j\omega-\beta_i}{j\omega+\overline{\beta}_i}\right|d\omega. \tag{8b}$$

The functions within the second and third integrals of (8b) are vanishing. Finally,

$$\int_{-\infty}^{\infty}\log|g(j\omega)|\frac{\sigma_k}{\sigma_k^2+(\omega-\eta_k)^2}d\omega=-\pi(-\log\left|g(\sigma_k+j\eta_k)\right|+\sum_{1\le i\le M}\log\left|\frac{\sigma_k+j\eta_k-\alpha_i}{\sigma_k+j\eta_k+\overline{\alpha}_i}\right|+\sum_{1\le i\le N}\log\left|\frac{\sigma_k+j\eta_k+\overline{\beta}_i}{\sigma_k+j\eta_k-\beta_i}\right|). \tag{9}$$

where the subscript k is associated with the arbitrary singular point. After embedding $s_0=\sigma_k+j\eta_k=\varsigma_k$, and $s_1=-\sigma_k+j\eta_k=-\overline{\varsigma}_k$ and ς_k is a kth NMP open-loop zero, where $1\le k\le L$, Eq. (9) boils down to

$$\int_{-\infty}^{\infty}\log|g(j\omega)|\frac{\sigma_k}{\sigma_k^2+(\omega-\eta_k)^2}d\omega=-\pi(\sum_{1\le i\le M}\log\left|\frac{\sigma_k+j\eta_k-\alpha_i}{\sigma_k+j\eta_k+\overline{\alpha}_i}\right|+\sum_{1\le i\le N}\log\left|\frac{\sigma_k+j\eta_k+\overline{\beta}_i}{\sigma_k+j\eta_k-\beta_i}\right|). \tag{10a}$$

For the case, the singular point coincides with the NMP open-loop zero, then the number of sensitivity integrals assuming the structure of (10a) would be the number of NMP open-loop zeros.

Suppose the sensitivity function $g(s)$ has no RHP poles, i.e., no RHP poles associated with the linear feedback system. The resulting sensitivity integral equation becomes

$$\int_{-\infty}^{\infty}\log|g(j\omega)|\frac{\sigma_k}{\sigma_k^2+(\omega-\eta_k)^2}d\omega=-\pi\sum_{1\le i\le M}\log\left|\frac{\sigma_k+j\eta_k-\alpha_i}{\sigma_k+j\eta_k+\overline{\alpha}_i}\right|. \quad \square \tag{10b}$$

Remark 1. In real-world practical scenarios, the feedback system does not exhibit good frequency response beyond a certain bandwidth due to unmodeled dynamics, nonlinearities and process parameter variations [8]. Hence, the above sensitivity integral (10b) can be decomposed into two integrals, i.e.

$$\int_0^{\omega_a}\log|g(j\omega)|\frac{\sigma_k}{\sigma_k^2+\omega^2}d\omega+\underset{\varepsilon\to 0}{\mathrm{Lim}}\int_{\omega_a+\varepsilon}^{\infty}\log|g(j\omega)|\frac{\sigma_k}{\sigma_k^2+\omega^2}d\omega=-\frac{\pi}{2}\sum_{1\le i\le M}\log\left|\frac{\sigma_k+j\eta_k-\alpha_i}{\sigma_k+j\eta_k+\overline{\alpha}_i}\right|, \tag{11}$$

where ω_a is the available bandwidth. The above (11) can be recast in a convenient setting that gives greater insights into the bandwidth limitation and augmentation of the concerning feedback system. Thus,

$$\int_0^{\omega_a} \log|g(j\omega)| \frac{\sigma_k}{\sigma_k^2 + (\omega - \eta_k)^2} d\omega = -\frac{\pi}{2} \sum_{1\le i\le M} \log\left|\frac{\sigma_k + j\eta_k - \alpha_i}{\sigma_k + j\eta_k + \overline{\alpha}_i}\right| + \delta(\omega_a),$$

$$\delta(\omega_a) = \lim_{\varepsilon\to 0} \int_{\omega_a+\varepsilon}^{\infty} \log|g(j\omega)| \frac{\sigma_k}{\sigma_k^2 + \omega^2} d\omega.$$

Due to this limitation in the feedback design, sensitivity improvement and deteriorations occur over the range $0 \le \omega \le \omega_a$. Here, we briefly explain two properties of the bandwidth perturbation parameter $\delta(\omega_a)$. Relatively less value of the parameter $\delta(\omega_a)$, the first choice, implies that the sensitivity integral for the frequency range $0 \le \omega \le \omega_a$ approaches closely to the sensitivity integral for the frequency $\omega \ge 0$. On the other hand, the larger value of the parameter choice $\delta(\omega_a)$, the second choice, introduces greater deviation into the sensitivity integral for the frequency range $0 \le \omega \le \omega_a$. Thus, the first choice of the parameter design results in the expanded bandwidth operation of the feedback system, the second choice of the parameter results in the limited bandwidth operation of the feedback system.

Remark 2. The result of Theorem 1 is a meaningful extension, motivated from practical reasons, of the result outlined in an older but seminal result of [9]. Theorem 1 accounts for the RHP poles of the sensitivity function, i.e. RHP poles of the feedback system leading to the instability. Greater detail about the instability in the feedback system in practical situations can be found in [35]. For the sake of generality, i.e. a theoretical perspective, Theorem 1 has non-trivial special cases as well. Interestingly, after combining the non-analytic function theory, not accounting for the singular point in the closed path as well as Cauchy residue integration, we achieve the Bode terminology of the sensitivity function.

Here, we rephrase the sensitivity integral of Theorem 1 in Bode's sensitivity integral equation setting. In the Bode integral setting, the singular point is not accounted for the closed path integral. On the other hand, the Poisson–Jensen integral equation accounts for the same. Since the Bode sensitivity integral is not a direct consequence of the Poisson–Jensen integral equation, we explain it succinctly. Recall the properties of the sensitivity functions $g(s)$ and $\widetilde{g}(s)$ of Theorem 1 of this paper; we rephrase the sensitivity integral equation, i.e.,

$$0 = \oint (\log g(s) + \sum_{1\le i\le M} \log\frac{s+\alpha_i}{s-\alpha_i} + \sum_{1\le i\le N} \log\frac{s-\beta_i}{s+\beta_i}) ds. \tag{12}$$

Eq. (12) is a consequence of two results, i.e., the Green closed path integral and the analyticity. Furthermore, the closed path integral can be regarded as integral in the open RHP in the control framework. The closed path integral can be decomposed as

$$\begin{aligned}
0 &= j\int_{-\infty}^{\infty} \log\, g(j\omega) d\omega + \underset{R\to\infty}{Lt} \int_{\frac{\pi}{2}}^{-\frac{\pi}{2}} \log(g(\mathrm{Re}^{j\gamma}))\mathrm{Re}^{j\gamma} j d\gamma + \sum_{1\le i\le M} j\int_{-\infty}^{\infty} \log\frac{j\omega+\alpha_i}{j\omega-\alpha_i} d\omega \\
&\quad + \sum_{1\le i\le M} \underset{R\to\infty}{Lt} \int_{\frac{\pi}{2}}^{-\frac{\pi}{2}} \log(\frac{\mathrm{Re}^{j\gamma}+\alpha_i}{\mathrm{Re}^{j\gamma}-\alpha_i}) j\mathrm{Re}^{j\gamma} d\gamma + \sum_{1\le i\le N} j\int_{-\infty}^{\infty} \log\frac{j\omega-\beta_i}{j\omega+\beta_i} d\omega + \sum_{1\le i\le N} \underset{R\to\infty}{Lt} \int_{\frac{\pi}{2}}^{-\frac{\pi}{2}} \log(\frac{\mathrm{Re}^{j\gamma}-\beta_i}{\mathrm{Re}^{j\gamma}+\beta_i}) j\mathrm{Re}^{j\gamma} d\gamma \\
&= j\int_{-\infty}^{\infty} \log\, g(j\omega) d\omega + \underset{R\to\infty}{Lt} \int_{\frac{\pi}{2}}^{-\frac{\pi}{2}} \log(1 + G(\mathrm{Re}^{j\gamma}))\mathrm{Re}^{j\gamma} j d\gamma + \sum_{1\le i\le M} j\int_{-\infty}^{\infty} \log\frac{j\omega+\alpha_i}{j\omega-\alpha_i} d\omega \\
&\quad + \sum_{1\le i\le M} \underset{R\to\infty}{Lt} \int_{\frac{\pi}{2}}^{-\frac{\pi}{2}} \log(\frac{\mathrm{Re}^{j\gamma}+\alpha_i}{\mathrm{Re}^{j\gamma}-\alpha_i}) j\mathrm{Re}^{j\gamma} d\gamma + \sum_{1\le i\le N} j\int_{-\infty}^{\infty} \log\frac{j\omega-\beta_i}{j\omega+\beta_i} d\omega + \sum_{1\le i\le N} \underset{R\to\infty}{Lt} \int_{\frac{\pi}{2}}^{-\frac{\pi}{2}} \log(\frac{\mathrm{Re}^{j\gamma}-\beta_i}{\mathrm{Re}^{j\gamma}+\beta_i}) j\mathrm{Re}^{j\gamma} d\gamma.
\end{aligned} \tag{13}$$

The poles of the sensitivity $g(s)$ function are attributed to the variable proportional gain a of the feedback system. For a given bound on the variable gain, the RHP pole arises. The second term of the right-hand side accounts for the open-loop transfer function $G(s)$ with the variable gain a and the unit relative degree, otherwise vanishes. Furthermore,

$$-a\pi = \underset{R\to\infty}{Lt} \int_{\frac{\pi}{2}}^{-\frac{\pi}{2}} \log(1 + G(\mathrm{Re}^{j\gamma}))\mathrm{Re}^{j\gamma} j d\gamma, \sum_{1\le i\le M} \int_{-\infty}^{\infty} \log\frac{j\omega+\alpha_i}{j\omega-\alpha_i} d\omega = 0, \sum_{1\le i\le N} \int_{-\infty}^{\infty} \log\frac{j\omega-\beta_i}{j\omega+\beta_i} d\omega = 0,$$

$$-2\pi j \sum_i \alpha_i = \sum_{1\le i\le M} \underset{R\to\infty}{Lt} \int_{\frac{\pi}{2}}^{-\frac{\pi}{2}} \log(\frac{\mathrm{Re}^{j\gamma}+\alpha_i}{\mathrm{Re}^{j\gamma}-\alpha_i}) j\mathrm{Re}^{j\gamma} d\gamma,$$

$$2\pi j \sum_i \beta_i = \sum_{1\le i\le M} \underset{R\to\infty}{Lt} \int_{\frac{\pi}{2}}^{-\frac{\pi}{2}} \log(\frac{\mathrm{Re}^{j\gamma}-\beta_i}{\mathrm{Re}^{j\gamma}+\beta_i}) j\mathrm{Re}^{j\gamma} d\gamma.$$

After plugging the above set of relations into (1), the concerning result in the Bode terminology becomes

$$\frac{-a\pi}{2} + \pi \sum_{1\le i\le M} \alpha_i - \pi \sum_{1\le i\le N} \beta_i = \int_0^{\infty} \log\,|g(j\omega)|\, d\omega. \tag{14}$$

The first term of the left-hand side accounts for the open-loop transfer function $G(s)$ with the variable gain a and the unit relative degree. For the case of 'other than the unit relative degree and the absence of the first term of the left-hand side', the integral reduces to the Bode integral of [8]. Eqs. (9) and (14) are two different sensitivity integral equations. Their usefulness,

Table 1
Sensitivity integrals for various cases of systems.

Case	Sensitivity integrals		
	Poisson–Jensen $\int_{-\infty}^{\infty} \log\lvert g(j\omega)\rvert \frac{\sigma_k}{\sigma_k^2+(\omega-\eta_k)^2} d\omega =$	Bode (relative degree = 1) $\int_0^{\infty} \log\lvert g(j\omega)\rvert\, d\omega =$	Bode (relative degree > 1) $\int_0^{\infty} \log\lvert g(j\omega)\rvert\, d\omega =$
1	$-\pi(\sum_{1\le i\le M} \log\left\lvert\frac{\sigma_k+j\eta_k-\alpha_i}{\sigma_k+j\eta_k+\overline{\alpha}_i}\right\rvert + \sum_{1\le i\le N} \log\left\lvert\frac{\sigma_k+j\eta_k+\overline{\beta}_i}{\sigma_k+j\eta_k-\beta_i}\right\rvert)$	$-\frac{a\pi}{2}+\pi\sum_{1\le i\le M}\alpha_i-\pi\sum_{1\le i\le N}\beta_i$	$\pi\sum_{1\le i\le M}\alpha_i-\pi\sum_{1\le i\le N}\beta_i$
2	$-\pi\sum_{1\le i\le M} \log\left\lvert\frac{\sigma_k+j\eta_k-\alpha_i}{\sigma_k+j\eta_k+\overline{\alpha}_i}\right\rvert$	$-\frac{a\pi}{2}+\pi\sum_{1\le i\le M}\alpha_i$	$\pi\sum_{1\le i\le M}\alpha_i$ [8]
3	0	$-\frac{a\pi}{2}$	0 [8]

i.e., the Poisson–Jensen and Bode, can be studied by considering appealing practical problems. For the NMP open-loop zero case, associated with the open-loop transfer function $G(s)$, the sensitivity integral equation assumes an appealing structure, see (10b). This setup is a convenient form to compute sensitivity inequality.

The first term of the left-hand side of (14) accounts for the open-loop transfer function $G(s)$ with the variable gain a and the unit relative degree. For the case of 'other than the unit relative degree and the absence of the third term of the right-hand side', the integral reduces to the Bode integral of [8], which is depicted in Table 1 given below.

Considering the NMP zeros and RHP poles in the sensitivity function $g(s)$, there arise three cases, i.e. *Case 1* — NMP zeros and RHP poles present, *Case 2* — NMP zeros present and RHP poles absent, *Case 3* — NMP zeros and RHP poles absent. Table 1 displays the associated Poisson–Jensen as well as Bode sensitivity integrals for all three special cases.

Note that the Poison–Jensen integral (Eq. (10a)) accounts for the singular point; on the other hand, the Bode integral does not account. Moreover, in the Poisson–Jensen setting, because of the association of the singular point in the denominator polynomial of sensitivity function, the relative degree does not decide the sensitivity integral; whereas, in Bode's integral, the relative degree decides.

Theorem 2. *Suppose the sensitivity function obeys the properties of Theorem 1 of this paper,* $\rho = \min_{0\le\omega<\omega_c} |g(j\omega)| \in (0,1)$, *and* $s_{\max} = \max_{\omega_c<\omega} |g(j\omega)| \in (1,\infty)$. *Then, the sensitivity inequalities are*

$$\log s_{\max} \ge \frac{\left|\frac{\pi}{2}(-\log|g(\sigma_k)| + \sum_{1\le i\le M}\log\left|\frac{\sigma_k-\alpha_i}{\sigma_k+\overline{\alpha}_i}\right| + \sum_{1\le i\le N}\log\left|\frac{\sigma_k+\overline{\beta}_i}{\sigma_k+\beta_i}\right|)\right| - |\log\rho|\tan^{-1}\frac{\omega_c}{\sigma_k}}{(\frac{\pi}{2}-\tan^{-1}\frac{\omega_c}{\sigma_k})},$$

where the singular point is the real-valued as well as contained in the right-half plane. Furthermore

$$\log s_{\max} \ge \frac{\left|\frac{\pi}{2}(\sum_{1\le i\le M}\log\left|\frac{\sigma_k-\alpha_i}{\sigma_k+\overline{\alpha}_i}\right| + \sum_{1\le i\le N}\log\left|\frac{\sigma_k+\overline{\beta}_i}{\sigma_k+\beta_i}\right|)\right| - |\log\rho|\tan^{-1}\frac{\omega_c}{\sigma_k}}{(\frac{\pi}{2}-\tan^{-1}\frac{\omega_c}{\sigma_k})},$$

where the singular point is the real-valued, contained in the right-half plane as well as coincides with an NMP open-loop zero.

Proof. Consider the Poisson–Jensen sensitivity integral

$$\int_{-\infty}^{\infty}\log|g(j\omega)|\frac{\sigma_k}{\sigma_k^2+(\omega-\eta_k)^2}d\omega = -\pi(-\log|g(\sigma_k+j\eta_k)| + \sum_{1\le i\le M}\log\left|\frac{\sigma_k+j\eta_k-\alpha_i}{\sigma_k+j\eta_k+\overline{\alpha}_i}\right| + \sum_{1\le i\le N}\log\left|\frac{\sigma_k+j\eta_k+\overline{\beta}_i}{\sigma_k+j\eta_k+\beta_i}\right|). \tag{15}$$

Note that the integrand on the left-hand side of (11) is not an even function. To achieve a simplified and appealing form of the sensitivity integral, consider that the singular points are the real. Then, the integrand is an even function. As a result of this,

$$\int_0^{\infty}\log|g(j\omega)|\frac{\sigma_k}{\sigma_k^2+\omega^2}d\omega = -\frac{\pi}{2}(-\log|g(\sigma_k)| + \sum_{1\le i\le M}\log\left|\frac{\sigma_k-\alpha_i}{\sigma_k+\overline{\alpha}_i}\right| + \sum_{1\le i\le N}\log\left|\frac{\sigma_k+\overline{\beta}_i}{\sigma_k+\beta_i}\right|).$$

After rephrasing in the sense of absolute values, the above becomes

$$\begin{aligned}\left|\int_0^{\infty}\log|g(j\omega)|\frac{\sigma_k}{\sigma_k^2+\omega^2}d\omega\right| &= \left|\int_0^{\omega_c}\log|g(j\omega)|\frac{\sigma_k}{\sigma_k^2+\omega^2}d\omega + \int_{\omega_c}^{\infty}\log|g(j\omega)|\frac{\sigma_k}{\sigma_k^2+\omega^2}d\omega\right| \\ &= \left|-\frac{\pi}{2}(-\log|g(\sigma_k)| + \sum_{1\le i\le M}\log\left|\frac{\sigma_k-\alpha_i}{\sigma_k+\overline{\alpha}_i}\right| + \sum_{1\le i\le N}\log\left|\frac{\sigma_k+\overline{\beta}_i}{\sigma_k+\beta_i}\right|)\right| \\ &= \left|\frac{\pi}{2}(-\log|g(\sigma_k)| + \sum_{1\le i\le M}\log\left|\frac{\sigma_k-\alpha_i}{\sigma_k+\overline{\alpha}_i}\right| + \sum_{1\le i\le N}\log\left|\frac{\sigma_k+\overline{\beta}_i}{\sigma_k+\beta_i}\right|)\right|.\end{aligned} \tag{16}$$

As a consequence of the triangular inequality, we have

$$\left|\int_0^{\omega_c} \log|g(j\omega)| \frac{\sigma_k}{\sigma_k^2+\omega^2} d\omega + \int_{\omega_c}^{\infty} \log|g(j\omega)| \frac{\sigma_k}{\sigma_k^2+\omega^2} d\omega\right| \leq \left|\int_0^{\omega_c} \log|g(j\omega)| \frac{\sigma_k}{\sigma_k^2+\omega^2} d\omega\right| + \left|\int_{\omega_c}^{\infty} \log|g(j\omega)| \frac{\sigma_k}{\sigma_k^2+\omega^2} d\omega\right|. \tag{17}$$

Note that

$$\max_{0\leq\omega<\omega_c} |\log|g(j\omega)|| = \left|\log \min_{0\leq\omega<\omega_c} |g(j\omega)|\right| = |\log\rho|.$$

After invoking the triangular inequality into the first term of the right-hand side of (17) and combining with the above relation, we have

$$\left|\int_0^{\omega_c} \log|g(j\omega)| \frac{\sigma_k}{\sigma_k^2+\omega^2} d\omega\right| \leq \int_0^{\omega_c} |\log|g(j\omega)|| \frac{\sigma_k}{\sigma_k^2+\omega^2} d\omega \leq |\log\rho| \tan^{-1}\frac{\omega_c}{\sigma_k}. \tag{18}$$

After adopting the appropriate notations, the following holds

$$\max_{\omega_c<\omega} |\log|g(j\omega)|| = \log \max_{\omega_c<\omega} |g(j\omega)| = \log s_{\max}$$

After combining the above relation with the last term of the right-hand side of (17), we have

$$\left|\int_{\omega_c}^{\infty} \log|g(j\omega)| \frac{\sigma_k}{\sigma_k^2+\omega^2} d\omega\right| = \int_{\omega_c}^{\infty} \log|g(j\omega)| \frac{\sigma_k}{\sigma_k^2+\omega^2} d\omega \leq (\frac{\pi}{2} - \tan^{-1}\frac{\omega_c}{\sigma_k}) \log s_{\max}. \tag{19}$$

The inequality (17) in conjunction with (18) and (19) boils down to

$$\begin{aligned}\left|\int_0^{\omega_c} \log|g(j\omega)| \frac{\sigma_k}{\sigma_k^2+\omega^2} d\omega + \int_{\omega_c}^{\infty} \log|g(j\omega)| \frac{\sigma_k}{\sigma_k^2+\omega^2} d\omega\right| &\leq \left|\int_0^{\omega_c} \log|g(j\omega)| \frac{\sigma_k}{\sigma_k^2+\omega^2} d\omega\right| + \left|\int_{\omega_c}^{\infty} \log|g(j\omega)| \frac{\sigma_k}{\sigma_k^2+\omega^2} d\omega\right| \\ &\leq |\log\rho| \tan^{-1}\frac{\omega_c}{\sigma_k} + (\frac{\pi}{2} - \tan^{-1}\frac{\omega_c}{\sigma_k}) \log s_{\max}.\end{aligned}$$

The above inequality in conjunction with Eq. (16) boils down to

$$\begin{aligned}\left|\int_0^{\infty} \log|g(j\omega)| \frac{\sigma_k}{\sigma_k^2+\omega^2} d\omega\right| &= \left|\frac{\pi}{2}(-\log|g(\sigma_k)| + \sum_{1\leq i\leq M} \log\left|\frac{\sigma_k-\alpha_i}{\sigma_k+\overline{\alpha}_i}\right| + \sum_{1\leq i\leq N} \log\left|\frac{\sigma_k+\overline{\beta}_i}{\sigma_k+\beta_i}\right|)\right| \\ &\leq |\log\rho| \tan^{-1}\frac{\omega_c}{\sigma_k} + (\frac{\pi}{2} - \tan^{-1}\frac{\omega_c}{\sigma_k}) \log s_{\max}.\end{aligned}$$

Alternatively

$$(\frac{\pi}{2} - \tan^{-1}\frac{\omega_c}{\sigma_k}) \log s_{\max} \geq \left|\frac{\pi}{2}(-\log|g(\sigma_k)| + \sum_{1\leq i\leq M} \log\left|\frac{\sigma_k-\alpha_i}{\sigma_k+\overline{\alpha}_i}\right| + \sum_{1\leq i\leq N} \log\left|\frac{\sigma_k+\overline{\beta}_i}{\sigma_k+\beta_i}\right|)\right| - |\log\rho| \tan^{-1}\frac{\omega_c}{\sigma_k}.$$

Finally, we have

$$\log s_{\max} \geq \frac{\left|\frac{\pi}{2}(-\log|g(\sigma_k)| + \sum_{1\leq i\leq M} \log\left|\frac{\sigma_k-\alpha_i}{\sigma_k+\overline{\alpha}_i}\right| + \sum_{1\leq i\leq N} \log\left|\frac{\sigma_k+\overline{\beta}_i}{\sigma_k+\beta_i}\right|)\right| - |\log\rho| \tan^{-1}\frac{\omega_c}{\sigma_k}}{(\frac{\pi}{2} - \tan^{-1}\frac{\omega_c}{\sigma_k})}. \tag{20}$$

Furthermore, sensitivity inequality

$$\log s_{\max} \geq \frac{\left|\frac{\pi}{2}(-\log|g(\sigma_k)| + \sum_{1\leq i\leq M} \log\left|\frac{\sigma_k-\alpha_i}{\sigma_k+\overline{\alpha}_i}\right|)\right| - |\log\rho| \tan^{-1}\frac{\omega_c}{\sigma_k}}{(\frac{\pi}{2} - \tan^{-1}\frac{\omega_c}{\sigma_k})}, \tag{21}$$

where the singular point is the real-valued, contained in the right-hand plane, and no RHP closed-loop poles are associated with the feedback system. □

Remark 3. The sensitivity inequality (21) is restrictive and a special case of (20). That is useful to achieve the sensitivity inequality for the stable linear feedback system, where the system has unstable open-loop poles. The inequality (21) further simplifies the case, i.e. the singular point coincides with the open-loop NMP zero. As a result of this, the term $\log|g(\sigma_k)|$ of (21) vanishes. Sensitivity inequality (20) covers a broader class of linear feedback systems. The restriction on the location of the sensitivity function's poles and zeros decides the sensitivity inequality. The sensitivity equality holds for frequency-independent transfer characteristics. On the other hand, equality will not hold for frequency-dependent transfer characteristics. The sensitivity inequality of Theorem 2 reveals the fact that logarithmic inequality arises from linear feedback systems.

2.2. Modified sensitivity integrals and related inequalities

The problem of improving the loop robustness reduces to the problem of reframing sensitivity bounds. The system's sensitivity gets affected due to controller failure in the case of process uncertainty, input disturbances, and limitations on control components.

The issue of losing the stability arises due to unavoidable performance limitations caused by unstable system open-loop dynamics and limitations of available bandwidth associated with the entities such as actuators, sensors, time-delays and control processors [8]. The area of amplification may spread over an extensive frequency range with only negligible amplification. In some cases, systems with narrow bandwidth actuators, where the region is squeezed into a narrow range of frequencies. That may lead to large peaks in the sensitivity function leading to strongly amplified disturbances for the closed-loop system and ultimately instability [35]. Popular examples are Chernobyl nuclear reactor, JAS-39 aircraft [8], unstable combustor control problem [36,37] and cavity oscillations [38,39].

In this section, we adopt the notion of the modified sensitivity to answer the problem of stabilizing the feedback system in the sensitivity inequality setting. This augments the ability of a linear feedback system to handle process parameter variations as well as input disturbances. To achieve that, we unify the idea of the Poisson–Jensen-type complex function integral with the modified sensitivity function. The modified sensitivity integrals and related inequalities are summarized in Theorems 3 and 4 of this paper.

Theorem 3. *Consider the sensitivity function $g(s)$ obeys the properties of Theorem 1. The modified sensitivity function $\widetilde{g}(s)$ is analytic, $\widetilde{g}(s) = g(s)\prod_{1\le i\le M}(\frac{s+\overline{\alpha}_i}{s-\alpha_i})\prod_{1\le i\le N}(\frac{s-\beta_i}{s+\overline{\beta}_i})$. Then, the following hold:*
(i) the Poisson–Jensen formula becomes

$$\int_{-\infty}^{\infty}\log\left|\widetilde{g}(j\omega)\right|\frac{\sigma_k}{\sigma_k^2+(\omega-\eta_k)^2}d\omega=\pi\log\left|\widetilde{g}(\sigma_k+j\eta_k)\right|,$$

where $\sigma_k + j\eta_k$ is an arbitrary singular point in the RHP.
(ii) $\int_{-\infty}^{\infty}\log\left|\widetilde{g}(j\omega)\right|\frac{\sigma_k}{\sigma_k^2+(\omega-\eta_k)^2}d\omega=0, 1\le k\le L.$
for the open-loop transfer function $\widetilde{G}(s)$ associated with the sensitivity function $\widetilde{g}(s)$, where the singular point coincides with the NMP open-loop zeros of $\widetilde{G}(s)$.
(iii) Suppose the singular point $\sigma_k + j\eta_k = \varsigma_k$, then

$$\int_{-\infty}^{\infty}\log\left|\widetilde{g}(j\omega)\right|\frac{\sigma_k}{\sigma_k^2+(\omega-\eta_k)^2}d\omega=\pi\log\left|\widetilde{g}(\sigma_k+j\eta_k)\right|, 1\le k\le S.$$

Note that ς_k is an NMP zero associated with the open-loop transfer $G(s)$ instead of the open-loop transfer function $\widetilde{G}(s)$. The integers L and S depend on the structure of $\widetilde{G}(s)$ and $G(s)$ respectively.

Proof. Consider 'the modified sensitivity function-sensitivity function relation'

$$\widetilde{g}(s)=\kappa(s)g(s), \kappa(s)=\prod_{1\le i\le M}(\frac{s+\overline{\alpha}_i}{s-\alpha_i})\prod_{1\le i\le N}(\frac{s-\beta_i}{s+\overline{\beta}_i}).$$

Furthermore,

$$\frac{1}{1+\widetilde{G}(s)}=\frac{\prod_{1\le i\le M}(\frac{s+\overline{\alpha}_i}{s-\alpha_i})\prod_{1\le i\le N}(\frac{s-\beta_i}{s+\overline{\beta}_i}).}{1+G(s)}, \widetilde{G}(s)=\frac{1-\kappa(s)+G(s)}{\kappa(s)}. \tag{22}$$

Recall equation (2b), we write

$$\begin{aligned}\log\widetilde{g}(s_0)&=-\frac{1}{2\pi j}\oint\frac{\log\widetilde{g}(s)}{s-s_0}ds+\frac{1}{2\pi j}\oint\frac{\log\widetilde{g}(s)}{s-s_1}\\&=-\frac{1}{2\pi j}\int_{-\infty}^{\infty}\frac{j(s_0-s_1)\log\widetilde{g}(j\omega)}{(j\omega-s_0)(j\omega-s_1)}d\omega-\frac{1}{2\pi j}\underset{R\to\infty}{Lt}\int_{\frac{\pi}{2}}^{-\frac{\pi}{2}}\frac{(s_0-s_1)\log\widetilde{g}(\mathrm{Re}^{j\gamma})}{(\mathrm{Re}^{j\gamma}-s_0)(\mathrm{Re}^{j\lambda}-s_1)}\mathrm{Re}^{j\gamma}jd\gamma\\&=-\frac{1}{2\pi j}\int_{-\infty}^{\infty}\frac{j(s_0-s_1)\log\widetilde{g}(j\omega)}{(j\omega-s_0)(j\omega-s_1)}d\omega.\end{aligned} \tag{23}$$

Note that

$$\underset{R\to\infty}{Lt}\frac{\log\widetilde{g}(\mathrm{Re}^{j\gamma})}{(\mathrm{Re}^{j\gamma}-s_0)(\mathrm{Re}^{j\gamma}-s_1)}\mathrm{Re}^{j\gamma}=\underset{R\to\infty}{Lt}\frac{-\log(1+\widetilde{G}(\mathrm{Re}^{j\gamma}))}{(\mathrm{Re}^{j\gamma}-s_0)(\mathrm{Re}^{j\gamma}-s_1)}\mathrm{Re}^{j\gamma}=0,$$

for every relative degree of the open-loop transfer function $\widetilde{G}(s)$. Here, we consider $s_0=\sigma_k+j\eta_k, s_1=-\sigma_k+j\eta_k$ as well as account for the real parts of both sides of (23). As a result of this, we are led to a more straightforward expression, i.e.,

$$\log\left|\widetilde{g}(\sigma_k+j\eta_k)\right|=\frac{1}{\pi}\int_{-\infty}^{\infty}\frac{\sigma_k\log\left|\widetilde{g}(j\omega)\right|}{\sigma_k^2+(\omega-\eta_k)^2}d\omega. \tag{24}$$

Consider the open-loop transfer function $\widetilde{G}(s)$ having NMP zeros, say the kth NMP zero of $\widetilde{G}(s)$ is ς_k. Suppose the singular point $\sigma_k + j\eta_k = \varsigma_k$, then (24) can be recast as

$$\int_{-\infty}^{\infty}\log\left|\widetilde{g}(j\omega)\right|\frac{\sigma_k}{\sigma_k^2+(\omega-\eta_k)^2}d\omega=0, 1\le k\le L.$$

The term $\log\ \widetilde{g}(j\omega)$ associated with Theorem 3 of this paper is analytic in the right half-plane. On the other hand, the term $\log\ g(j\omega)$ associated with Theorem 1 is non-analytic. Similar arguments can be made to arrive at the third sensitivity integral of Theorem 3 of this paper.

Remark 4. Eq. (22), in combination with the sensitivity integrals of Theorem 3, answers the problem of stabilizing the linear feedback system to handle input disturbances as well as process parameter variations. 'Reshaping the forward path transfer function' using (22) and 'reshaping the sensitivity inequality' can be achieved using the results of Theorem 3, which is sketched in Theorem 4 of this paper.

Theorem 4. *Suppose the sensitivity function obeys the properties of Theorem 3 of this paper,* $\rho = \min_{0\le\omega<\omega_c} \left|\widetilde{g}(j\omega)\right| \in (0,1)$, *and* $s_{\max} = \max_{\omega_c<\omega} \left|\widetilde{g}(j\omega)\right| \in (1,\infty)$. *Then, the sensitivity inequalities are*

$$\log s_{\max} \ge \frac{\frac{\pi}{2}\left|\log\left|\widetilde{g}(\sigma_k)\right|\right| - \left|\log\rho\right|\tan^{-1}\frac{\omega_c}{\sigma_k}}{(\frac{\pi}{2} - \tan^{-1}\frac{\omega_c}{\sigma_k})},$$

where the singular point is the real-valued, contained in the RHP. Furthermore,

$$\log s_{\max} \ge \frac{-\left|\log\rho\right|\tan^{-1}\frac{\omega_c}{\sigma_k}}{(\frac{\pi}{2} - \tan^{-1}\frac{\omega_c}{\sigma_k})},$$

where the singular point is the real-valued, contained in the RHP, and the singular point is the k*th NMP zero of the* $\widetilde{G}(s)$.

Proof. Recall equation (24), the concerning integral is

$$\int_{-\infty}^{\infty} \log\left|\widetilde{g}(j\omega)\right| \frac{\sigma_k}{\sigma_k^2 + (\omega-\eta_k)^2} d\omega = \pi \log\left|\widetilde{g}(\sigma_k + j\eta_k)\right|.$$

Consider the singular points are real, then the integrand in (24) is an even function. As a result of this, we arrive at

$$\int_{0}^{\infty} \log\left|\widetilde{g}(j\omega)\right| \frac{\sigma_k}{\sigma_k^2 + \omega^2} d\omega = \frac{\pi}{2} \log\left|\widetilde{g}(\sigma_k)\right|. \tag{25}$$

Since the closed-form expression of the logarithmic absolute sensitivity satisfying the integral (25) is not possible, we introduce the notion of the 'logarithmic inequality'. After rephrasing in the sense of absolute values, the above becomes

$$\left|\int_{0}^{\infty} \log\left|\widetilde{g}(j\omega)\right| \frac{\sigma_k}{\sigma_k^2 + \omega^2} d\omega\right| = \left|\int_{0}^{\omega_c} \log\left|\widetilde{g}(j\omega)\right| \frac{\sigma_k}{\sigma_k^2 + \omega^2} d\omega + \int_{\omega_c}^{\infty} \log\left|\widetilde{g}(j\omega)\right| \frac{\sigma_k}{\sigma_k^2 + \omega^2} d\omega\right| = \frac{\pi}{2}\left|\log\left|\widetilde{g}(\sigma_k)\right|\right|. \tag{26}$$

As a consequence of the triangular inequality, we have

$$\left|\int_{0}^{\omega_c} \log\left|\widetilde{g}(j\omega)\right| \frac{\sigma_k}{\sigma_k^2 + \omega^2} d\omega + \int_{\omega_c}^{\infty} \log\left|\widetilde{g}(j\omega)\right| \frac{\sigma_k}{\sigma_k^2 + \omega^2} d\omega\right| \le \left|\int_{0}^{\omega_c} \log\left|\widetilde{g}(j\omega)\right| \frac{\sigma_k}{\sigma_k^2 + \omega^2} d\omega\right| + \left|\int_{\omega_c}^{\infty} \log\left|\widetilde{g}(j\omega)\right| \frac{\sigma_k}{\sigma_k^2 + \omega^2} d\omega\right|. \tag{27}$$

Note that

$$\max_{0\le\omega<\omega_c} \left|\log\left|\widetilde{g}(j\omega)\right|\right| = \left|\log \min_{0\le\omega<\omega_c} \left|\widetilde{g}(j\omega)\right|\right| = \left|\log\rho\right|.$$

After invoking the triangular inequality into the first term of the right-hand side of (27) and then combining with the above relation, we have

$$\left|\int_{0}^{\omega_c} \log\left|\widetilde{g}(j\omega)\right| \frac{\sigma_k}{\sigma_k^2 + \omega^2} d\omega\right| \le \int_{0}^{\omega_c} \left|\log\left|\widetilde{g}(j\omega)\right|\right| \frac{\sigma_k}{\sigma_k^2 + \omega^2} d\omega \le \left|\log\rho\right|\tan^{-1}\frac{\omega_c}{\sigma_k}. \tag{28}$$

After adopting appropriate notations, the following holds

$$\max_{\omega_c<\omega} \left|\log\left|\widetilde{g}(j\omega)\right|\right| = \log \max_{\omega_c<\omega} \left|\widetilde{g}(j\omega)\right| = \log s_{\max}.$$

After combining the above relation with the last term of the right-hand side of (27), we have

$$\left|\int_{\omega_c}^{\infty} \log\left|\widetilde{g}(j\omega)\right| \frac{\sigma_k}{\sigma_k^2 + \omega^2} d\omega\right| = \int_{\omega_c}^{\infty} \log\left|\widetilde{g}(j\omega)\right| \frac{\sigma_k}{\sigma_k^2 + \omega^2} d\omega \le (\frac{\pi}{2} - \tan^{-1}\frac{\omega_c}{\sigma_k}) \log s_{\max}. \tag{29}$$

The inequality (27) in conjunction with (28) and (29) boils down to

$$\begin{aligned}\left|\int_{0}^{\omega_c} \log\left|\widetilde{g}(j\omega)\right| \frac{\sigma_k}{\sigma_k^2 + \omega^2} d\omega + \int_{\omega_c}^{\infty} \log\left|\widetilde{g}(j\omega)\right| \frac{\sigma_k}{\sigma_k^2 + \omega^2} d\omega\right| &\le \left|\int_{0}^{\omega_c} \log\left|\widetilde{g}(j\omega)\right| \frac{\sigma_k}{\sigma_k^2 + \omega^2} d\omega\right| + \left|\int_{\omega_c}^{\infty} \log\left|\widetilde{g}(j\omega)\right| \frac{\sigma_k}{\sigma_k^2 + \omega^2} d\omega\right| \\ &\le \left|\log\rho\right|\tan^{-1}\frac{\omega_c}{\sigma_k} + (\frac{\pi}{2} - \tan^{-1}\frac{\omega_c}{\sigma_k}) \log s_{\max}.\end{aligned}$$

The above inequality, in conjunction with Eq. (26), boils down to

$$\left|\int_0^\infty \log|\widetilde{g}(j\omega)|\,\frac{\sigma_k}{\sigma_k^2+\omega^2}d\omega\right| = \frac{\pi}{2}\,\big|\log|\widetilde{g}(\sigma_k)|\,\big| \le |\log\rho|\tan^{-1}\frac{\omega_c}{\sigma_k} + (\frac{\pi}{2}-\tan^{-1}\frac{\omega_c}{\sigma_k})\log s_{\max}.$$

Finally, we have

$$\log s_{\max} \ge \frac{\frac{\pi}{2}\,\big|\log|\widetilde{g}(\sigma_k)|\,\big| - |\log\rho|\tan^{-1}\frac{\omega_c}{\sigma_k}}{(\frac{\pi}{2}-\tan^{-1}\frac{\omega_c}{\sigma_k})}, \quad \square \tag{30}$$

Remark 5. It is worth mentioning that the logarithmic sensitivity inequality of the linear feedback system resembles the idea of the celebrated Cramer–Rao lower bound of signal estimations. The logarithmic sensitivity inequality is a lower bound inequality as well. A signal estimator associated with the minimum variance for a given set of n estimators is adjudged as the best signal estimator since that reveals the minimum fluctuation in the estimate. Interestingly, for an available set of n controllers for the given feedback system, a controller associated with the minimum $\log s_{\max}$, is adjudged as the best controller. The terms minimum and maximum are associated with a set of controllers and frequency variations, respectively. The controller with the minimum logarithmic sensitivity implies greater robustness ability to handle input disturbances and process parameter variations. The Appendix of this paper comments on the logarithmic lower bound sensitivity inequality.

3. Case study

From a design perspective, the use of a loop analysis tool is powerful. The impact of changes in the design of the controller $G_c(s)$ can be witnessed or measured if the inequalities are in terms of the loop transfer function. The efficacy of a controller can be measured in several terms. The major one is the robustness to process parameter variations and responses to reference signals and disturbances. The first step for a controller to provide adequate robustness performance is to satisfy the sensitivity inequalities. The sensitivity function represents the disturbance attenuation/amplification and also relates the uncertainties in the process parameter. In this paper, we consider a case study example and four established controllers for sensitivity analysis from the control viewpoint by adopting the sensitivity inequality framework of the Theorems of this paper. Consider a general class of Unstable Second-Order Plus Dead Time (USOPDT) system transfer function given by

$$G_p(s) = \frac{k_p e^{-t_d s}}{(\tau_1 s-1)(\tau_2 s+1)}. \tag{31}$$

Numerous practical problems possess the form of an open-loop unstable system, e.g., chemical reactors with the exothermic reaction [40], jacket-cooled CSTR [41], etc. For this reason, Eq. (31) is examined via the introduction of sensitivity integrals. In the logarithmic integral, the singular point coincides with the NMP open-loop zero. The exponential delay $e^{-t_d s}$ annihilates the contribution of the closed path sensitivity integral in the Poisson–Jensen's, excluding the integral along the imaginary axis. Consider a specific transfer function $G_p(s)$ with an unstable RHP pole [42,43], i.e.,

$$G_p(s) = \frac{e^{-0.939s}}{(5s-1)(2.07s+1)}. \tag{32}$$

Here, we consider four appealing controllers available in the literature [44–47] and examine them in a sensitivity inequality perspective. Their controller structure is

$$G_{cm}(s) = k_c(1+\frac{1}{\tau_I s}+\tau_D s),\ m=1 \text{ to } 4. \tag{33}$$

The sensitivity function for (33) in the forward path with (31) is given by

$$g_m(s) = \frac{s\tau_I(\tau_1 s-1)(\tau_2 s+1)}{s\tau_I(\tau_1 s-1)(\tau_2 s+1)+k_p k_c(1+\tau_I s+\tau_I\tau_D s^2)e^{-t_d s}},$$

$$g_m(j\omega) = \frac{\tau_I\omega^2(\tau_2-\tau_1)+j\tau_1\omega(1-\omega^2\tau_1\tau_2)}{\tau_I\omega^2(\tau_2-\tau_1)+j\tau_1\omega(1-\omega^2\tau_1\tau_2)+k_p k_c(1+j\omega\tau_I-\tau_I\tau_D\omega^2)e^{-jt_d\omega}}$$

$$= \frac{x_2+jy_2}{k_p k_c(x_3\cos t_d\omega+\tau_I\omega\sin t_d\omega)+x_2+j(k_p k_c(\tau_I\omega\cos t_d\omega-x_3\sin t_d\omega)+y_2)},$$

where $x_2=\tau_I\omega^2(\tau_2-\tau_1)$, $y_2=\tau_1\omega(1-\omega^2\tau_1\tau_2)$, $x_3=1-\tau_I\tau_D\omega^2$.

The controller efficacy is tested using the idea of sensitivity integrals and sensitivity inequalities for controller structure (33) with four different controller settings; see Fig. 2 depicts logarithmic absolute sensitivity. Table 2 unfolds the fact that the increase in $s_{\max}(m)$ is associated with the decrease in $\rho(m)$, where $1 \le m \le n$ and $n=4$. That implies the increase in the positive value of the term $|\log\rho(m)|$, since $\rho(m)$ is contained in the open internal $(0,1)$. The frequency term $\omega_c(m)$ increases with increase in the term $s_{\max}(m)$. As a result of this, the sensitivity lower bound term $s_{lb}(\omega_c(m),\sigma(m))$ increases, see Eq. (A.3) of Appendix as well. The numerical studies of Table 2 concerning a set of four controllers agree with a theoretical discussion of the Appendix; see Fig. 2 as well for the graphical representation of the discussion.

The less absolute maximum sensitivity reveals the higher stability margin. The higher margin preserves the robustness and brings apart from the fragility under the influence of input disturbances and process parameter variations. On the other hand, the

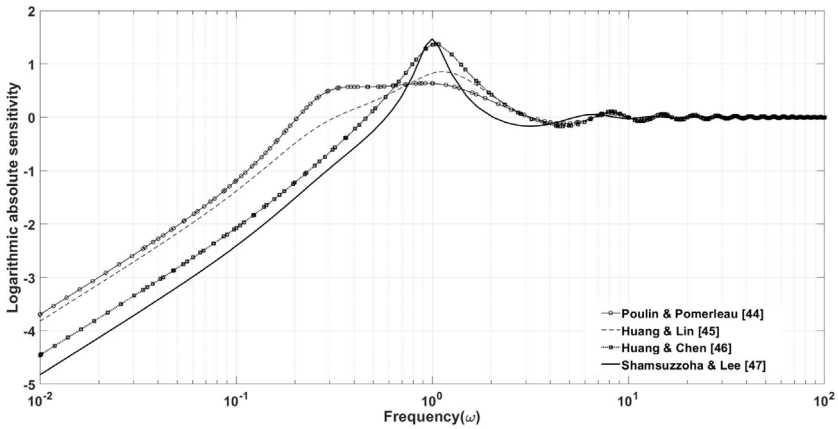

Fig. 2. Graphical representation of $\left|\log g_m(j\omega)\right|$.

Table 2
Sensitivity lower bounds and indices.

Controllers	Controller parameters			Sensitivity lower bound	Sensitivity indices			
	k_c	τ_I	τ_D	$s_{lb}(\omega_c(m), \sigma(m))$	$s_{\max}(m)$	$\rho(m)$	$\lvert\log \rho(m)\rvert$	$\omega_c(m)$
Poulin & Pomerleau [44]	3.95	4.95	2.07	0.2011	1.89	0.024838	3.6904	0.20281
Huang & Lin [45]	6.18	7.17	1.47	0.2083	2.36	0.021898	3.8214	0.31328
Huang & Chen [46]	3.05	7.55	2.07	0.2183	3.95	0.011602	4.4565	0.45583
Shamsuzzoha & Lee [47]	7.01	5.62	1.49	0.2277	4.34	0.008025	4.8251	0.58222

maximum value of $s_{\max} = 4.34$, associated with the controller of [47] indicates maximum amplification of disturbances occurring at the frequencies $\omega > \omega_c$. On increasing the uncertainty in the process parameter, the maximum sensitivity increases, leading to a decrease in the stability margin and ultimately to failure. For these reasons, the controller of [44] is recommended in the sensitivity context.

The computation of absolute maximum sensitivity function value requires algorithmic procedures as well as numerical simulations. On the other hand, concerning the sensitivity lower bound, the calculation is quite simple and direct. The minimum of sensitivity lower bound is associated with the minimum of absolute maximum sensitivity value. A more significant discussion can be found in Appendix. Thus, the sensitivity lower bound $s_{lb}(\omega_c(m), \sigma(m))$ suffices to examine the ability of controllers. That eliminates the computational subtleties associated with the term $s_{\max}$. Here, we outline a procedure for examining the ability of controllers to handle process uncertainties and disturbances.

(i) Determine the presence of NMP zeros and RHP poles in the sensitivity transfer function $g_m(j\omega)$, where $1 \leq m \leq n$ and n denotes the number of controllers. Note that the sensitivity NMP zero is the same as the associated RHP open-loop pole.
(ii) Calculate the frequency $\omega_c(m)$ that satisfies the condition $\left|g_m(j\omega_c)\right| = 1$.
(iii) Utilize Eq. (A.3) of the Appendix, and then calculate the controller-specific sensitivity lower bound. The controller associated with the minimum of sensitivity lower bound is adjudged with the greater ability to handle the effect of input disturbances as well as process parameter variations.

4. Conclusion

The main contribution of this paper is the extension of the conventional sensitivity integrals for linear feedback systems with non-analytic sensitivity functions in Poisson–Jensen's terminology. The sensitivity inequalities of this paper can be considered as useful techniques to assess the performance of controllers. Another contribution of this paper is the proposed simplified procedure for the effortless computation of sensitivity lower bounds without relying on simulation and complex computation. The loop robustness of linear feedback systems can be adjudged by eliminating the computational subtleties. In conjunction with the rearranged sensitivity lower bound for an unstable case study example, the theoretical results of this paper reveal the usefulness and applicability of the proposed method to examine and augment loop robustness. The technique of the paper is also suggestive of application to other appealing control systems.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix. Comments on the concerning sensitivity lower bound

We wish to examine the controllers' ability for a given plant in the feedback setting by combining the idea of sensitivity function and the Poisson–Jensen sensitivity inequality. The usefulness of the sensitivity inequality can be adjudged using the notion of computational tractability associated with relatively fewer calculations. The sensitivity inequality encompasses the calculation of the associated lower bound. Alternatively, the direct computation of the sensitivity function exploits the algorithmic procedure as well as numerical studies. To gain insights, we wish to weave a connection between the logarithmic sensitivity $\log s_{\max}$, and the associated sensitivity lower bound. After adopting appropriate notations, we have

$$\log s_{\max}(m) = \log \max_{\omega>\omega_c} \left|g_m(j\omega)\right| = \log \max_{0<\omega<\infty} \left|g_m(j\omega)\right|, \tag{A.1}$$

where m runs over 1 to n, n is the number of possible controllers. After considering the singular point is real-valued, contained in the right-half plane and coincides with the NMP open-loop zero of the system transfer function. As a result of this, the sensitivity inequality, a special case of (17), we have

$$\log s_{\max} \geq \frac{\left|\frac{\pi}{2}\left(\sum_{1\leq i\leq M} \log\left|\frac{\sigma_k-\alpha_i}{\sigma_k+\overline{\alpha}_i}\right| + \sum_{1\leq i\leq N} \log\left|\frac{\sigma_k+\overline{\beta}_i}{\sigma_k+\beta_i}\right|\right)\right| - |\log\rho|\tan^{-1}\frac{\omega_c}{\sigma_k}}{\left(\frac{\pi}{2} - \tan^{-1}\frac{\omega_c}{\sigma_k}\right)}.$$

To gain insights and unfold the lower bound property in the control perspective, consider the case sensitivity function has NMP zeros with the open-loop RHP poles. As a result of this, the logarithmic sensitivity inequality

$$\log s_{\max} \geq \frac{\left|\frac{\pi}{2}\sum_{1\leq i\leq M} \log\left|\frac{\sigma_k-\alpha_i}{\sigma_k+\overline{\alpha}_i}\right|\right| - |\log\rho|\tan^{-1}\frac{\omega_c}{\sigma_k}}{\left(\frac{\pi}{2} - \tan^{-1}\frac{\omega_c}{\sigma_k}\right)}.$$

Suppose a set of n possible controllers for a given plant, and the controllers do not contribute to the plant RHP poles and NMP zeros. This results in n sensitivity functions, where each has the same NMP zeros. Furthermore, the parameters ω_c and ρ are the controller-dependent. Thus, adopt the notations $\omega_c(m)$ and $\rho(m)$, where $1 \leq m \leq n$. After adopting the appropriate notations, the above structure becomes

$$\log s_{\max}(m) \geq \frac{\left|\frac{\pi}{2}\sum_{1\leq i\leq M} \log\left|\frac{\sigma_k-\alpha_i}{\sigma_k+\overline{\alpha}_i}\right|\right| - \left|\log\rho_m\right|\tan^{-1}\frac{\omega_c(m)}{\sigma_k}}{\left(\frac{\pi}{2} - \tan^{-1}\frac{\omega_c(m)}{\sigma_k}\right)}, \tag{A.2}$$

Alternatively,

$$\log s_{\max}(m) + \frac{\left|\log\rho_m\right|\tan^{-1}\frac{\omega_c(m)}{\sigma_k}}{\left(\frac{\pi}{2} - \tan^{-1}\frac{\omega_c(m)}{\sigma_k}\right)} \geq \frac{\left|\frac{\pi}{2}\sum_{1\leq i\leq M} \log\left|\frac{\sigma_k-\alpha_i}{\sigma_k+\overline{\alpha}_i}\right|\right|}{\left(\frac{\pi}{2} - \tan^{-1}\frac{\omega_c(m)}{\sigma_k}\right)},$$

where $1 \leq m \leq n$. Using the shorthand notations, the above is recast as

$$\log s_{\max}(m) + \frac{\left|\log\rho_m\right|\tan^{-1}\frac{\omega_c(m)}{\sigma_k}}{\left(\frac{\pi}{2} - \tan^{-1}\frac{\omega_c(m)}{\sigma_k}\right)} \geq s_{lb}(\omega_c(m), \sigma(m)), \tag{A.3}$$

where the right-hand side of the lower bound inequality (A.2) is $s_{lb}(\omega_c(m), \sigma(m))$. Now, we wish to unfold bound properties of (A.3):

(i) Notably, the above inequality is the controller-dependent as well as the controller does not contribute to the non-minimum phase property of the associated sensitivity function as well as the non-minimum phase property of the plant is the 'controller-independent. Thus, the absolute sensitivity function $\left|g_m(j\omega)\right|$, where $1 \leq m \leq n$, evolves differently but shows a similar increase–decrease over the frequency variation.

(ii) The increase in the term $\max_{0<\omega<\infty} \left|g_m(j\omega)\right|$ implies the increase in the logarithmic sensitivity $\log s_{\max}$, where m varies. That also increases the sensitivity lower bound $s_{lb}(\omega_c(m), \sigma(m))$, since the greater $\max_{0<\omega<\infty} \left|g_m(j\omega)\right|$ is associated with the greater $\left|\ln\rho_m\right|$ and the greater $\omega_c(m)$. As a result of this, the combined effect of the two terms of the left-hand side of the lower bound inequality shows the increasing property. We rephrase the lower bound property. Suppose a set of absolute maximum sensitivity values associated with a set of n controllers for a given plant. Consider the controllers' property restriction is that the controllers do not augment the NMP zero of the plant. Construct an increasing sequence of absolute maximum sensitivity values. The increasing sequence of the absolute maximum sensitivity values suggests the increasing sequence of the sensitivity lower bounds. The above properties suggest the usefulness of the sensitivity lower bound to choose a controller under the controller property restriction. The controller with $\min_{1\leq m\leq n} s_{lb}(\omega_c(m), \sigma(m))$, which implies $\min_{1\leq m\leq n} s_{\max}$, is the best robust controller. Thus, the computation of $\min_{1\leq m\leq n} s_{lb}(\omega_c(m), \sigma(m))$, an alternative method, suffices to choose a controller with relatively fewer efforts in contrast to the sensitivity computation $\min_{1\leq m\leq n}\left|g_m(j\omega)\right|$ via the algorithmic procedure.

References

[1] Åström KJ, Murray RM. Feedback systems: An introduction for scientists and engineers. New Jersey: Princeton University Press; 2008.
[2] Zames G, Francis BA. Minimax sensitivity, and optimal robustness. IEEE Trans Automat Control 1983;28:585–601.
[3] Boyd S, Deoser CA. Subharmonic functions and performance bounds on linear time-invariant feedback systems. IMA J Math Control Inform 1985;2:153–70.
[4] Jensen J. Sur un nouvel et important théorème de la théorie des fonctions. Acta Math 1899;22:359–64, (in French).
[5] Levinson N, Redheffer RM. Complex variables. San Francisco: Holden-Day; 1970.
[6] Fisher S. Introduction to complex variables. SIAM Rev 1971;13:409–11.
[7] Freudenberg JS, Looze DP. Sensitivity reduction, nonminimum phase zeros, and design trade-offs in single loop feedback systems. In: The 22nd IEEE conference on decision control. IEEE; 1983, p. 625–30.
[8] Stein G. Respect the unstable. IEEE Control Syst Mag 2003;23:12–25.
[9] Freudenberg JS, Looze DP. Right half plane poles and zeros and design trade-offs in feedback systems. IEEE Trans Automat Control 1985;30:555–65.
[10] Chen J. Sensitivity integral relations and design trade-offs in linear multivariable feedback systems. IEEE Trans Automat Control 1995;40:1700–16.
[11] Havre H, Skogestad S. Effect of RHP zeros and poles on the sensitivity functions in multivariable systems. J Process Control 2009;8:155–64.
[12] Yamamoto Y, Wakaiki W, Özbay H. Sensitivity reduction by stable controllers for MIMO infinite dimensional systems via the tangential Nevanlinna-Pick interpolation. IEEE Trans Automat Control 2014;59:1099–105.
[13] Iglesias PA. Trade-offs in linear time-varying systems: An analogue of Bode's sensitivity integral. Automatica 2001;37:1541–50.
[14] Sandberg H, Bernhardsson B. A bode sensitivity integral for linear time-periodic systems. IEEE Trans Automat Control 2005;50:2034–9.
[15] Karthikeyan R, Sinha S, Patil V. On the resolution of the sensitivity conjecture. Bull Amer Math Soc 2020;57:615–38.
[16] Saravanakumar T, Nirmala VJ, Raja R, Cao J, Lu G. Finite-time reliable dissipative control of neutral-type switched artificial neural networks with nonlinear fault inputs and randomly occurring uncertainties. Asian J Control 2019;22:2487–99.
[17] Zhu Q, Saravanakumar T, Gomathi S, Anthoni SM. Finite-time extended dissipative based optimal guaranteed cost resilient control for switched neutral systems with stochastic actuator failures. IEEE Access 2019;7:90289–303.
[18] Saravanakumar T, Anthoni SM, Zhu Q. Resilient extended dissipative control for Markovian jump systems with partially known transition probabilities under actuator saturation. J Franklin Inst 2020;357:6197–227.
[19] Saravanakumar T, Muoi NH, Zhu Q. Finite-time sampled-data control of switched stochastic model with non-deterministic actuator faults and saturation nonlinearity. J Franklin Inst 2020;357:13637–65.
[20] Dorosti M, Fey RHB, Heertjes MF, Nijimeijer H. Iterative Pole–Zero model updating: A combined sensitivity approach. Control Eng Pract 2018;71:164–74.
[21] Yu T, Zhao J, Xu Z, Chen X, Biegler LT. Sensitivity-based hierarchical distributed model predictive control of nonlinear processes. J Process Control 2019;84:146–67.
[22] Ringwood JV, Mérigaud A, Faedo N, Fusco F. An analytical and numerical sensitivity and robustness analysis of wave energy control systems. IEEE Trans Control Syst Technol 2020;28:1337–48.
[23] Alfaro VM, Vilanova R. Robust tuning of 2DoF five-parameter PID controllers for inverse response controlled processes. J Process Control 2013;23:453–62.
[24] Chu M, Xu C, Chu J. Graphic IMC-PID tuning based on maximum sensitivity for uncertain systems. Trans Inst Meas Control 2018;41:2196–204.
[25] Wang DJ, Li W, Guo ML. Tuning of $PI^{\lambda}D^{\mu}$ controllers based on sensitivity constraint. J Process Control 2013;23:861–7.
[26] Li D, Liu L, Jin Q, Hirasawa K. Maximum sensitivity based fractional IMC-PID controller design for non-integer order system with time delay. J Process Control 2015;31:17–29.
[27] Ruben M, Daniel S, Daniel N, Cesar DP. Coordination of distributed model predictive controllers using price-driven coordination and sensitivity analysis. In: 10th IFAC international symposium on dynamics and control of process systems. Elsevier; 2013, p. 215–20.
[28] Martín C, Bermúdez M, Barrero F, Arahal MR, Kestelyn X, Durán MJ. Sensitivity of predictive controllers to parameter variation in five-phase induction motor drives. Control Eng Pract 2017;68:23–31.
[29] Rani KY. Sensitivity compensating control: Data-driven model based adaptive approach. J Process Control 2011;21:1265–86.
[30] Xiong Y, Jing T, Chen T. Sensitivity analysis and sensitivity-based design for linear alarm filters. Control Eng Pract 2018;70:29–39.
[31] Åström KJ. Limitations on control system performance. Eur J Control 2000;6:2–20.
[32] Costa-Castelló R, Dormido S. An interactive tool to introduce the waterbed effect. IFAC-PapersOnLine 2015;48:259–64.
[33] Roberts GN, Towill DR. Sensitivity aspects of pre-compensator design for a ship-steering/ stabilization system. Trans Inst Meas Control 1988;10:33–6.
[34] Bernstein DS. What makes some control problems hard? IEEE Control Syst Mag 2002;22:8–19.
[35] Rowley CW, Batten BA. Dynamic and closed-loop control. In: Joslin RD, Miller DN, editors. Fundamentals and applications of modern flow control. Reston: American Institute of Aeronautics and Astronautics Inc; 2009, p. 115–48.
[36] Cohen JM, Banaszuk A. Factors affecting the control of unstable combustors. J Propuls Power 2003;19:811–21.
[37] Banaszuk A, Mehta PG, Jacobson CA, Khibnik AI. Limits of achievable performance of controlled combustion processes. IEEE Trans Control Syst Technol 2006;14:881–95.
[38] Rowley CW, Williams DR, Colonius T, Murray RM, Macmynowski DG. Linear models for control of cavity flow oscillations. J Fluid Mech 2006;547:317–30.
[39] Rowley CW, Williams DR. Dynamics and control of high-Reynolds-number flow over open cavities. Annu Rev Fluid Mech 2006;38:251–76.
[40] Luyben WL. Process modeling, simulation and control chemical engineers. 2nd ed.. New York: McGraw-Hill; 1990.
[41] Luyben WL. Tuning temperature controllers on openloop unstable reactors. Ind Eng Chem Res 1998;37:4322–31.
[42] Tan W, Marques HJ, Chen T. IMC design for unstable processes with time delays. J Process Control 2002;13:203–13.
[43] Yang XP, Wang QG, Hang CC, Lin C. IMC-based control system design for unstable processes. Ind Eng Chem Res 2002;41:4288–94.
[44] Poulin É, Pomerleau A. Tuning for integrating and unstable processes. IEE Proc Control Theory Appl 1996;143:429–35.
[45] Huang CT, Lin YS. Tuning PID controller for open-loop unstable processes with time delay. Chem Eng Commun 1995;133:11–30.
[46] Huang HP, Chen CC. Control-system synthesis for open-loop unstable process with time delay. IEE Proc Control Theory Appl 1997;144:334–46.
[47] Shamsuzzoha M, Lee M. IMC-PID controller design for improved disturbance rejection of time-delayed processes. Ind Eng Chem Res 2007;46:2077–91.